

## AL-2001

### Artificial Intelligence

### Lab # 7

#### Objectives:

- Operator Overriding, Operator Overloading

**Note:** Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

#### Problems:

Create a Python program for an advanced library management system that incorporates a **LibraryCatalog** class. This program demonstrates operator overloading for addition, subtraction, and equality, along with custom methods for length and string representation. The system enables the management of extensive library catalogs while providing in-depth functionality.

#### Class Definitions:

1. **Book Class:**
  - The **Book** class represents individual books within the library catalog.
  - It includes attributes such as **title**, **author**, **isbn**, **year**, and **price**.
2. **LibraryCatalog Class:**
  - The **LibraryCatalog** class serves as the core component of the library management system.
  - It has the following operator overloading and custom methods:

#### Operator Overloading:

- **\_\_add\_\_(self, other):** Overrides the addition operator to merge two library catalogs into a new one.
- **\_\_sub\_\_(self, other):** Overrides the subtraction operator to subtract the books of one catalog from another.

- **\_\_eq\_\_(self, other)**: Overrides the equality operator to compare two library catalogs for equality based on book attributes.

#### Custom Methods:

- **\_\_len\_\_(self)**: Calculates and returns the number of books in the catalog.
- **\_\_str\_\_(self)**: Provides a formatted string representation of the catalog, including book titles and relevant details.

#### Program Execution:

1. Create instances of the **Book** class to represent individual books within the library catalog.
2. Create instances of the **LibraryCatalog** class to build the library catalogs. Populate them with books.
3. Demonstrate the operator overloading for catalog addition, subtraction, and equality.
4. Calculate and display the number of books in a catalog using the custom **\_\_len\_\_** method.
5. Display a formatted string representation of the catalog using the custom **\_\_str\_\_** method.

#### Hints:

- Use the **\_\_add\_\_** method to merge catalogs and handle possible duplicates.
- Implement error handling in the **\_\_sub\_\_** method to ensure safe catalog subtraction.
- Ensure comprehensive book attribute comparisons in the **\_\_eq\_\_** method for accurate equality checks.
- Create separate methods for comparing book attributes to facilitate the **\_\_eq\_\_** method.
- Track the number of books in the catalog and update it when books are added or removed.
- Provide a user-friendly, human-readable representation of the catalog in the **\_\_str\_\_** method.

Question end hints started.

#### Hints for Individual operators

You are developing a comprehensive library management system, and you have a **LibraryCatalog** class that represents a collection of books. Overriding the **+** operator for **LibraryCatalog** objects enables you to combine two library catalogs seamlessly. This can be useful when merging the catalogs of two libraries or when adding new acquisitions to the existing catalog.

#### Hints:

- Consider creating a method, such as **merge\_catalog**, within the **LibraryCatalog** class to handle the addition of two catalogs.

- Think about how you'll ensure that duplicate books are not added multiple times.
- You may need to implement logic to handle the merging of book copies or editions.

In the same library management system, overriding the `-` operator for **LibraryCatalog** objects allows you to perform operations like catalog subtraction. This may be necessary when one library acquires another, and you need to subtract the books from the acquired library's catalog from the combined catalog. Additionally, it helps manage the removal of lost or damaged books from the catalog.

#### Hints:

- Consider creating a method, such as **subtract\_catalog**, within the **LibraryCatalog** class to handle catalog subtraction.
- Think about how you'll identify books that should be removed from the catalog.
- Implement proper error handling to ensure that you're not removing books that are not present in the catalog.

Your library management system involves federated libraries across multiple campuses. To ensure that two library catalogs are identical, you need to compare them for equality based on various attributes like book titles, authors, ISBNs, and publication years. Overriding the `==` operator for **LibraryCatalog** objects allows you to perform in-depth catalog comparisons and synchronize catalogs accurately.

#### Hints:

- Define a method, such as **compare\_catalogs**, within the **LibraryCatalog** class to perform the comparison.
- Consider implementing a comprehensive comparison logic that accounts for book attributes and metadata.
- Think about how you'll handle situations where books may have variations, such as different editions or translations.

Consider a situation where your library management system includes budget allocation based on the value of books in a catalog. To determine the catalog with higher book value, you may need to compare the catalogs using the `<` operator. This involves calculating the total value of books in each catalog and then comparing them.

#### Hints:

- Create a method, such as **compare\_value**, within the **LibraryCatalog** class to calculate the total value of books in the catalog.

- Implement logic to calculate the value of books, considering factors like book prices and the number of copies.
- Ensure that the comparison logic provides the expected results for budget allocation decisions.

Your library management system involves federated libraries across multiple campuses. To ensure that two library catalogs are identical, you need to compare them for equality based on various attributes like book titles, authors, ISBNs, and publication years. Overriding the `==` operator for **LibraryCatalog** objects allows you to perform in-depth catalog comparisons and synchronize catalogs accurately.

#### Hints:

- Create a method, such as **compare\_catalogs**, within the **LibraryCatalog** class to perform the comparison.
- Implement a comprehensive comparison logic that accounts for book attributes and metadata.
- Consider creating helper methods to compare specific attributes (e.g., title, author, ISBN) and then use these methods within the main comparison logic.
- Handle situations where books may have variations, such as different editions or translations, by considering additional attributes.

Your library management system needs to provide a way to calculate the total number of books in a **LibraryCatalog**. Implement the `__len__` method to return the count of books when `len(catalog)` is called.

#### Hints:

- Define the `__len__` method within the **LibraryCatalog** class to calculate and return the count of books.
- Consider using a class attribute or instance attribute to keep track of the number of books, and update it when books are added or removed.
- Ensure that the `__len__` method accurately reflects the current count of books in the catalog.



Users of your library management system want to obtain a user-friendly, human-readable representation of the **LibraryCatalog**. Implement the `__str__` method to provide a formatted string that displays the catalog's book titles and associated information.

**Hints:**

- Create the `__str__` method within the **LibraryCatalog** class to format the catalog's information.
- Iterate through the books in the catalog and construct a string that includes book titles and relevant details.
- Consider providing additional options for customizing the output format or limiting the number of displayed books for long catalogs.

**You need to do your exercise within the given time.**